AWS VPC PEERING

Project Completion & Evidence Report

Field	Details
Student name	Debi Prasad Nayak
Batch	Aws Internship Batch 1
Project	AWS VPC Peering — Private Connectivity
AWS Region	US East (N. Virginia) us-east-1
Completion date	29 October 2026

Project objective

Create two non-overlapping VPCs and configure VPC peering so two private EC2 workloads can communicate over private IPv4 addresses. A public bastion in VPC A provides administrative SSH access.

Evidence note: This report organizes the supplied screenshots captured during the lab. AWS resources were subsequently deleted as part of cleanup.

1. Architecture and resource plan

The workbook specifies VPC A (10.10.0.0/16) and VPC B (10.20.0.0/16), with a public bastion subnet in VPC A and private workload subnets in both VPCs. The peering connection carries private traffic; both sides require routes and appropriate security-group rules.

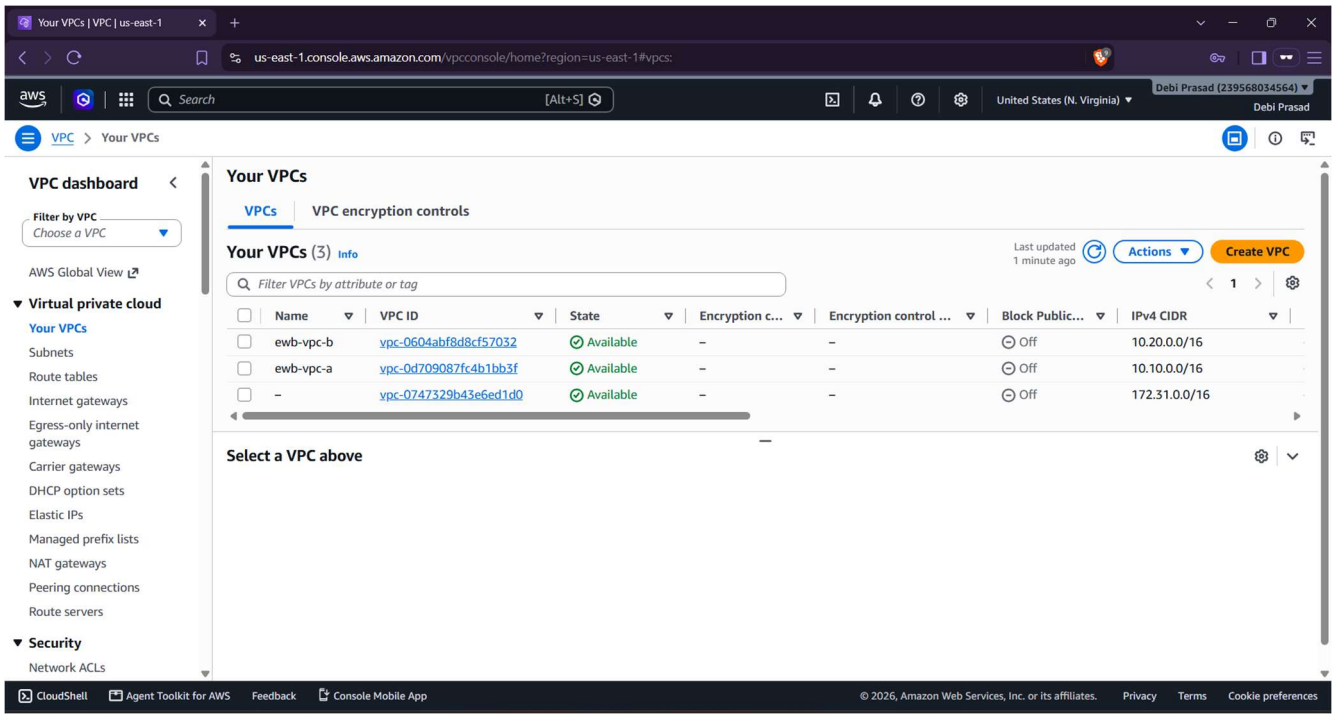
Resource	CIDR / address
ewb-vpc-a	10.10.0.0/16
ewb-a-public	10.10.1.0/24
ewb-a-private	10.10.2.0/24
ewb-vpc-b	10.20.0.0/16
ewb-b-private	10.20.1.0/24
Bastion	10.10.1.10 + public IPv4
App-A	10.10.2.10 (private)
App-B	10.20.1.10 (private)

Note: The supplied EC2 screenshots show the actual private IP addresses used in the lab. Where they differ from the workbook's example addresses, the actual values visible in the screenshots should be treated as the evidence of execution.

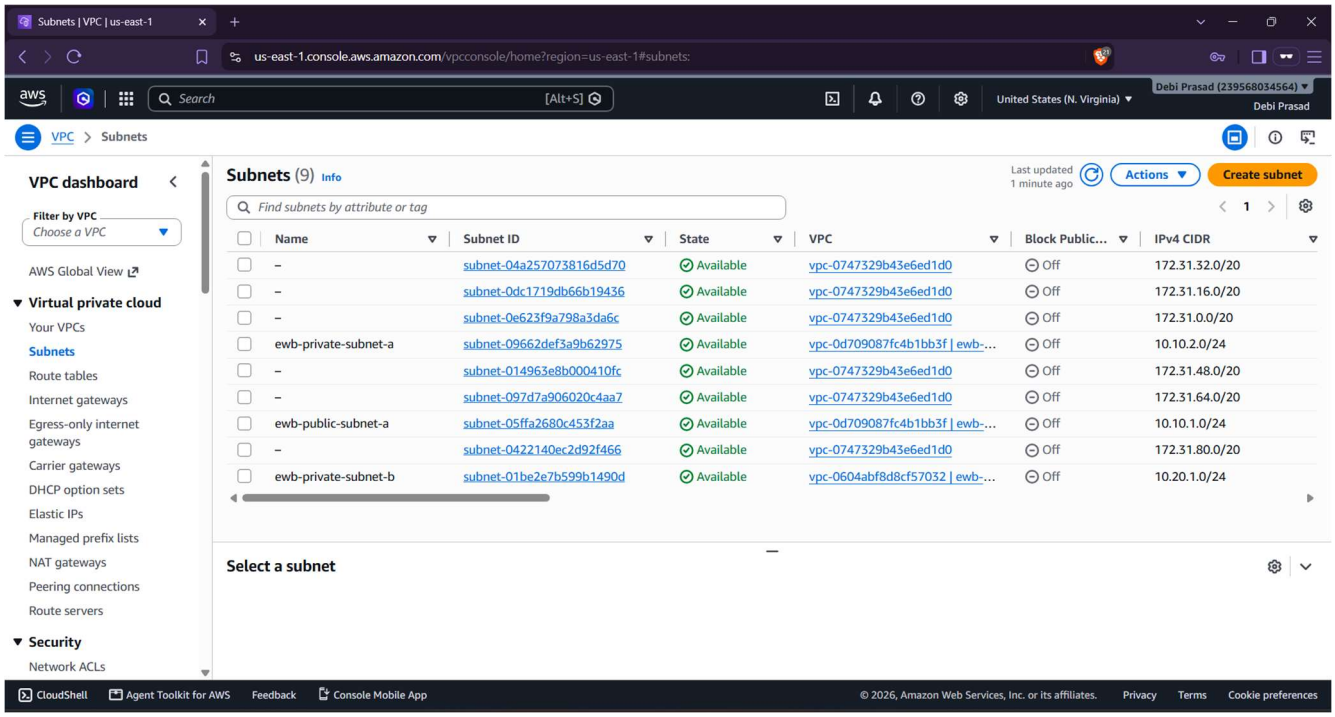
2. Network configuration: VPCs and subnets

The following console screenshots document VPC and subnet setup. Review them alongside the resource plan above.

VPC configuration



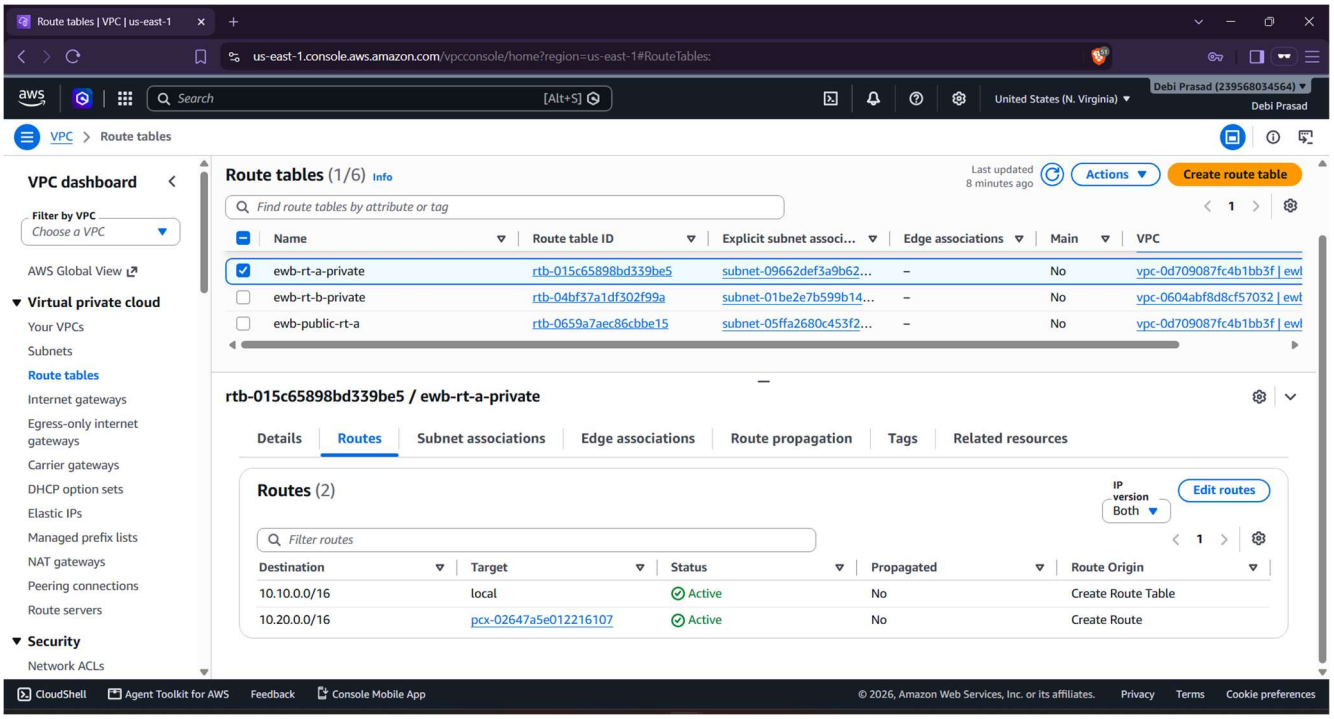
Subnet inventory



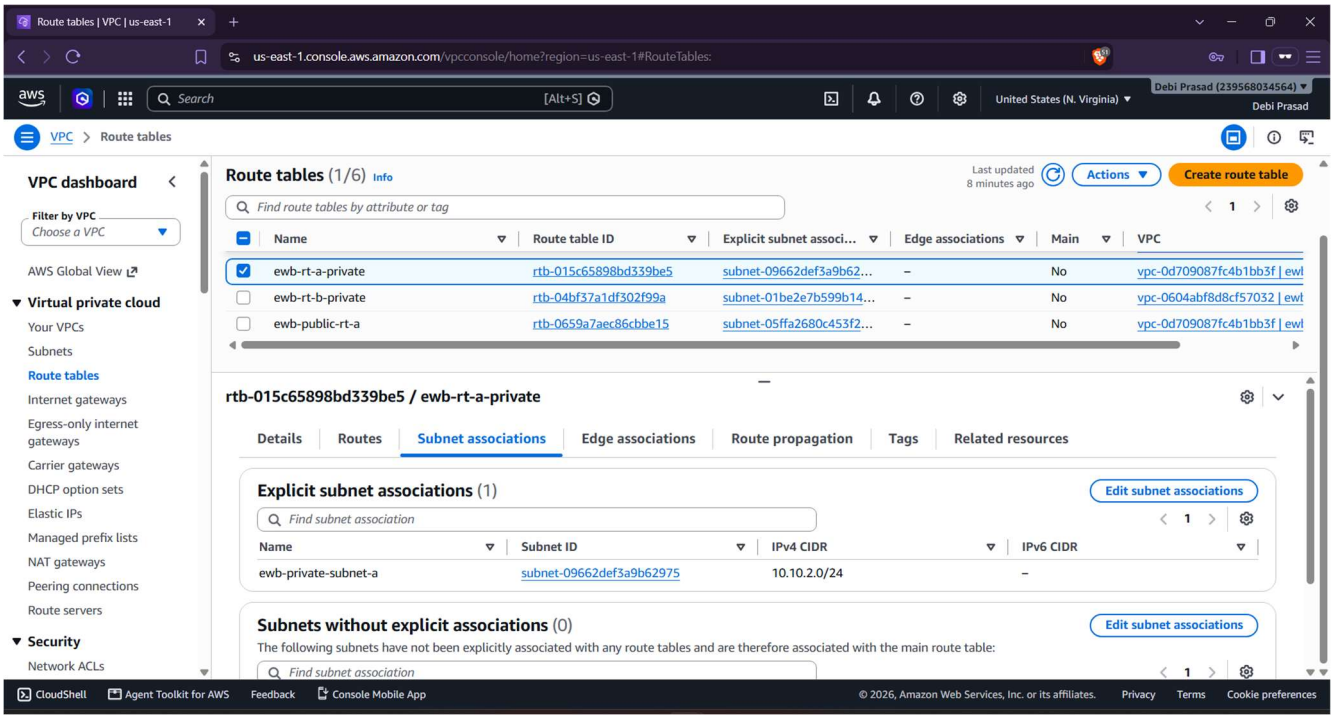
3. Route tables and associations

These screenshots document the route-table pages and route/association details captured during setup. The required design uses a peering route on both private route tables and a peering route on the public route table for bastion-to-App-B SSH.

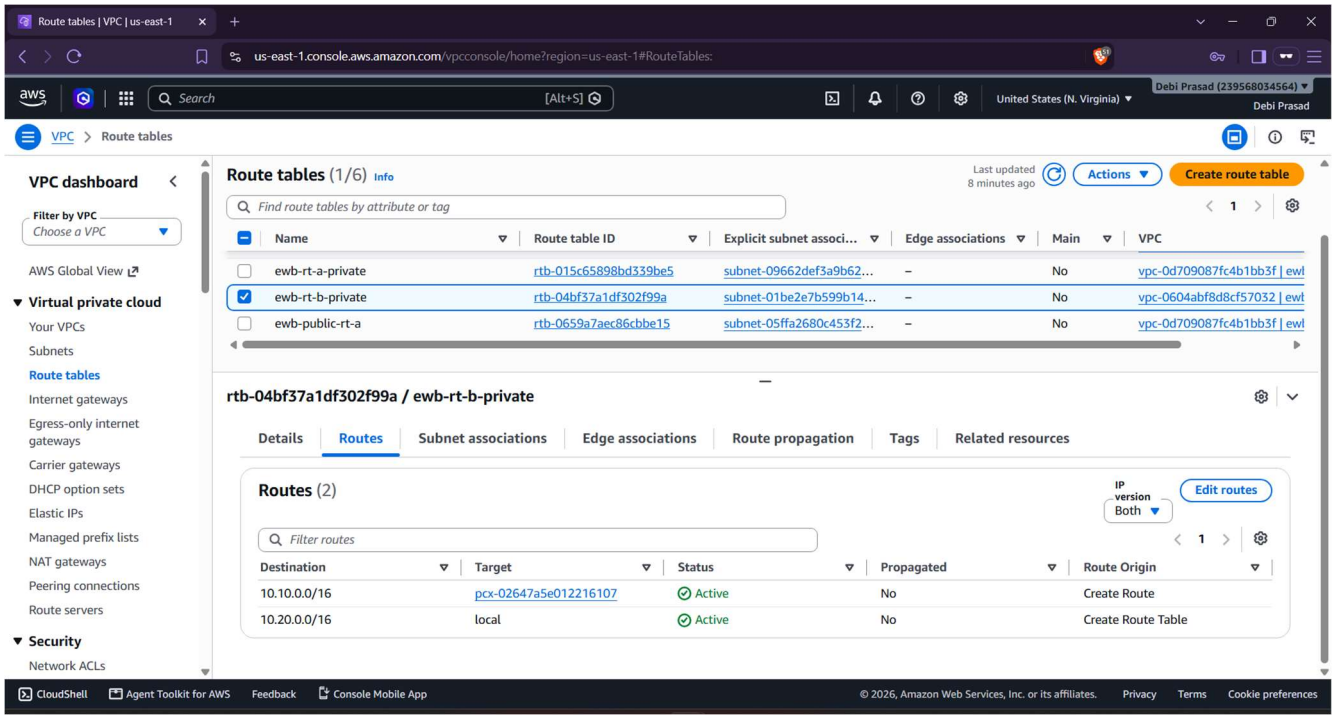
Route table configuration



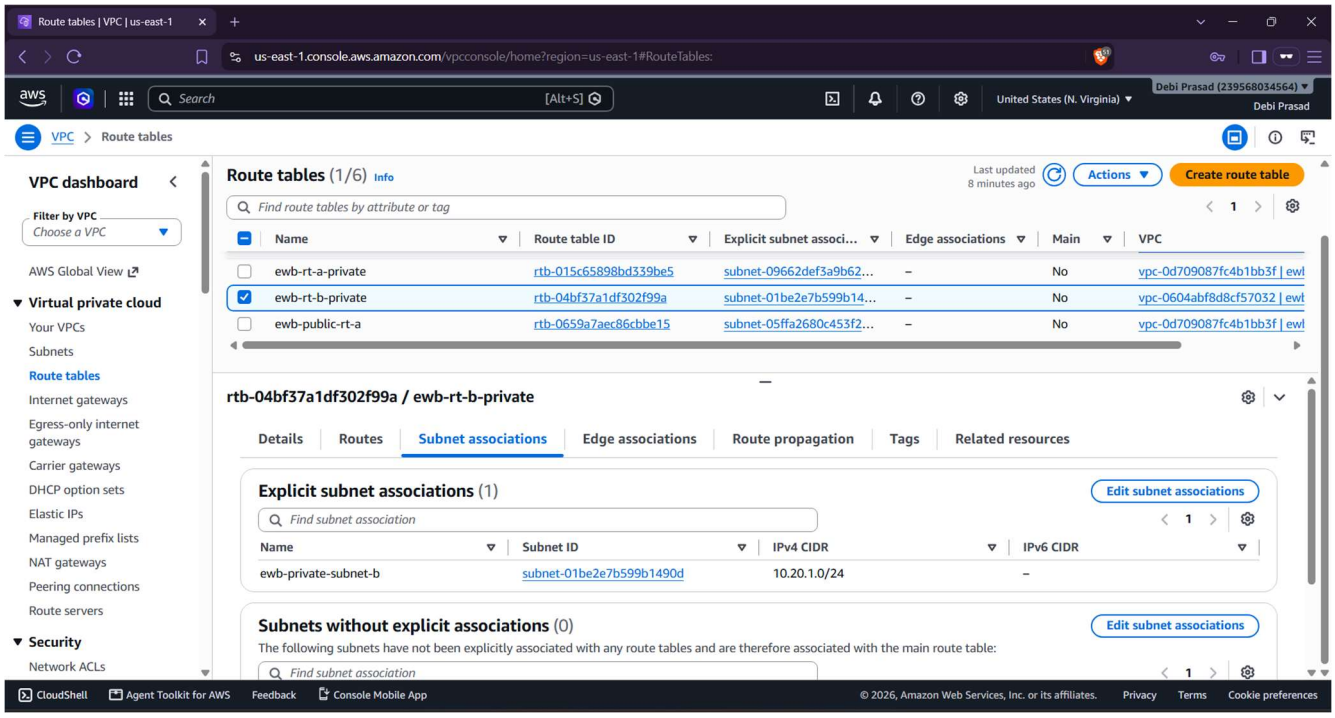
Route table details



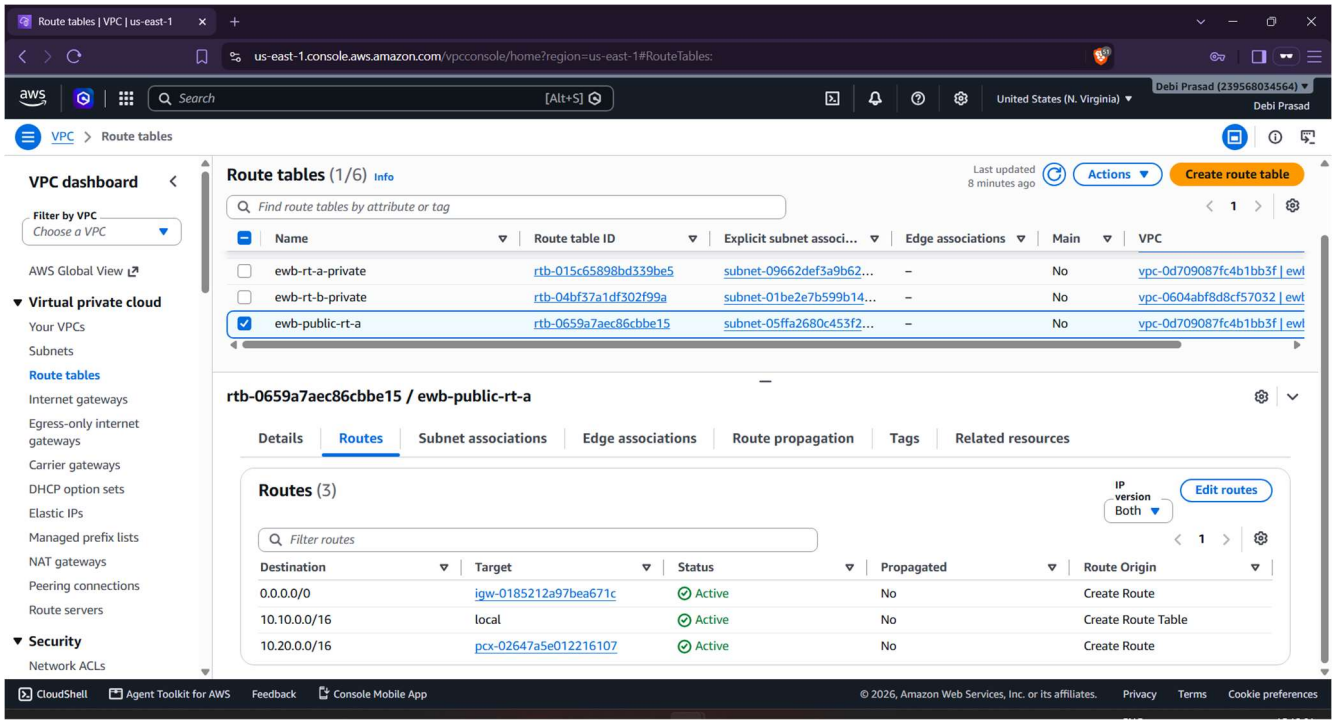
Route table routes and associations



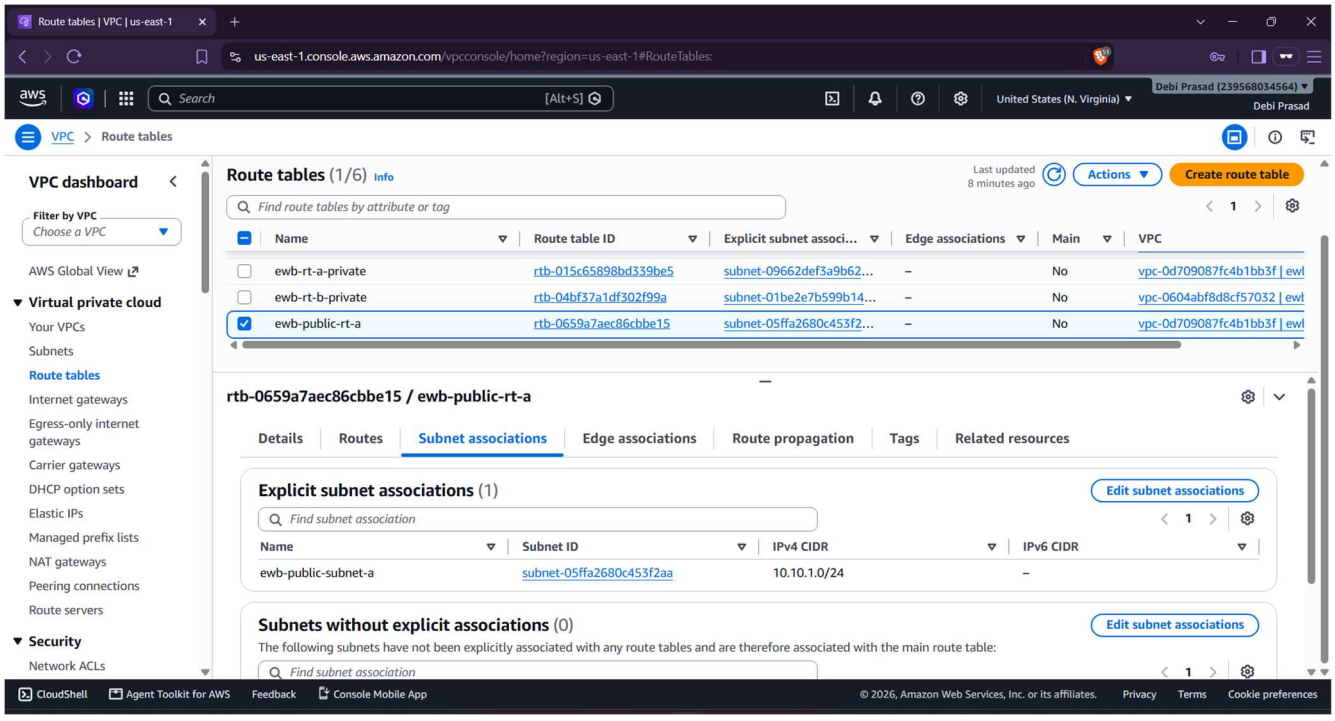
Route table details



Route table configuration



Route table details

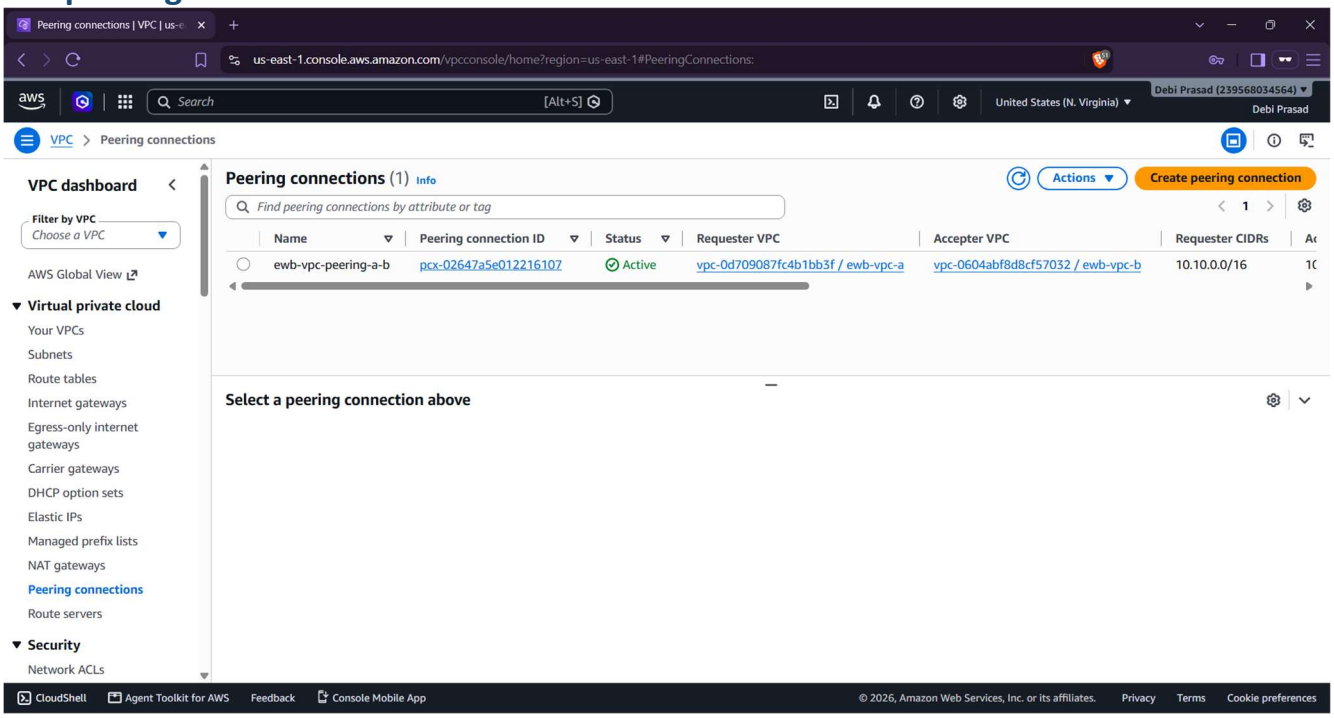


Original screenshot: Screenshot (2429).png

4. VPC peering connection

The screenshot below records the peering-connection console view captured during the lab. Confirm that the status and connection details are legible in the image.

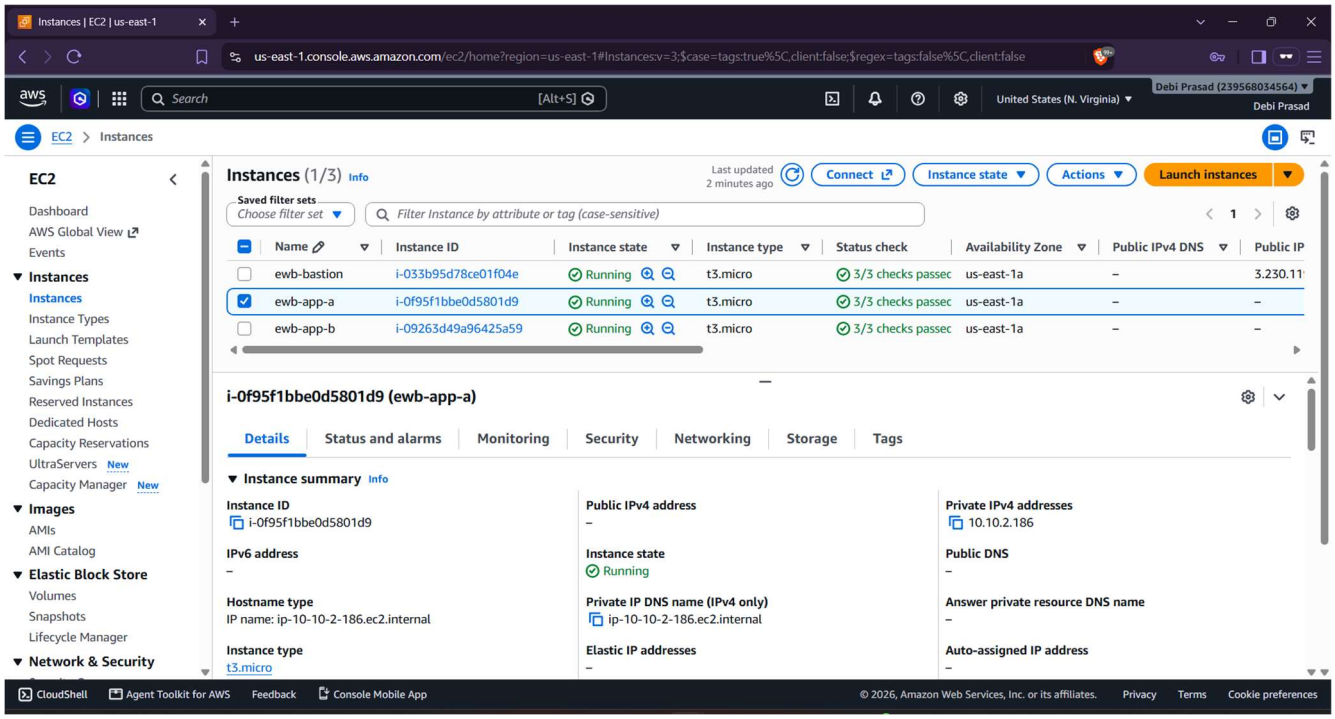
VPC peering connections



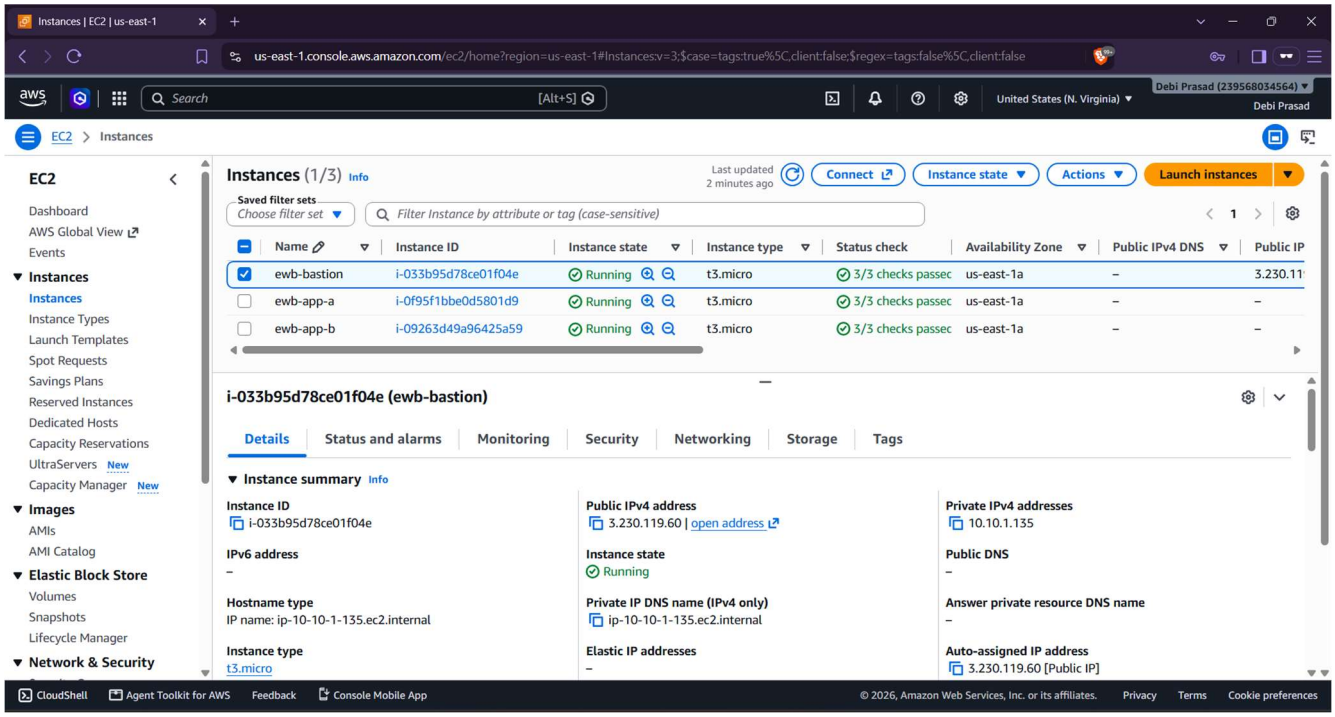
5. EC2 instances and security groups

The screenshots document the EC2 instance console, instance details and security-group inventory captured during the lab.

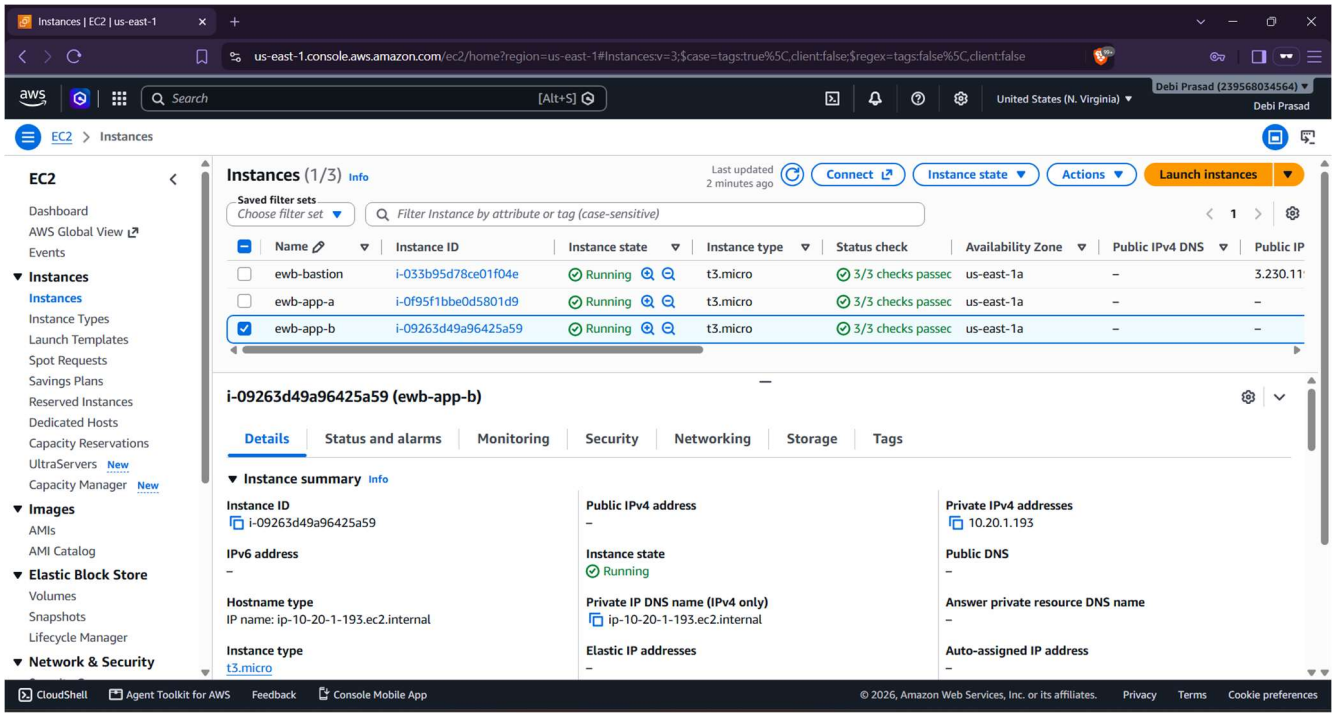
EC2 instances



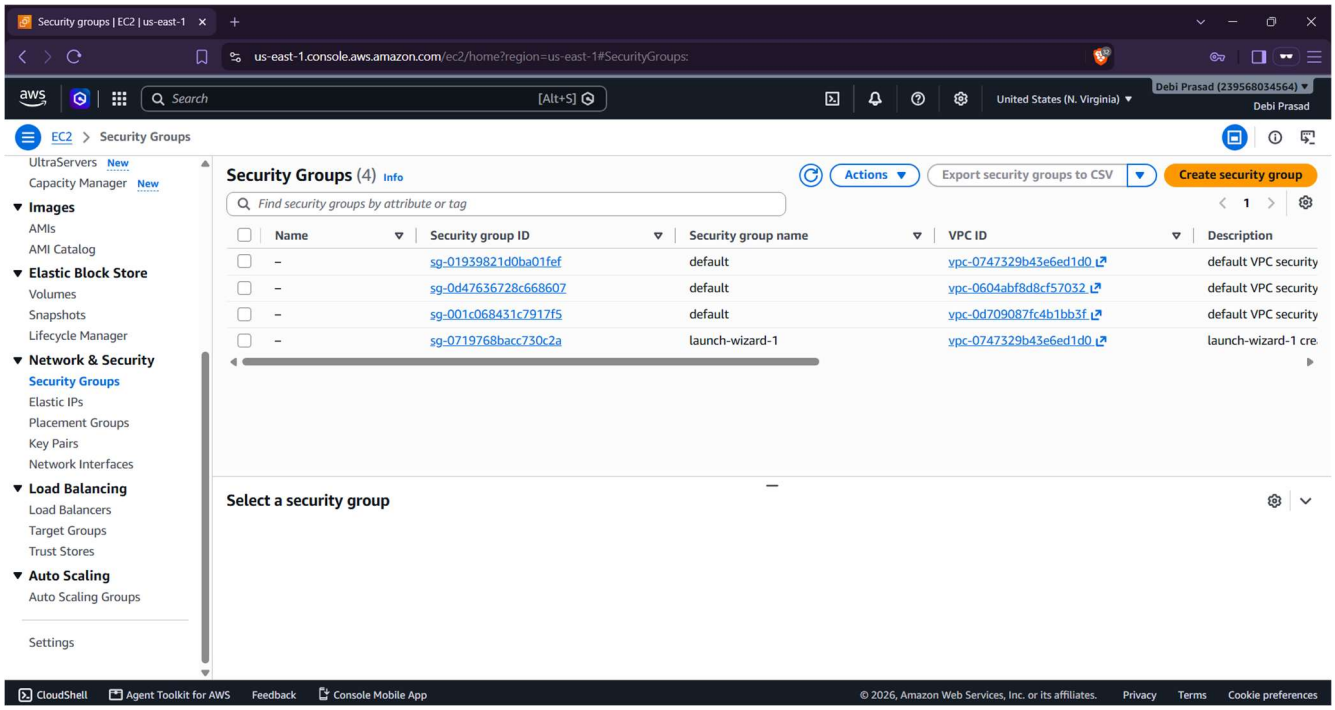
EC2 instance details



EC2 instance details and status



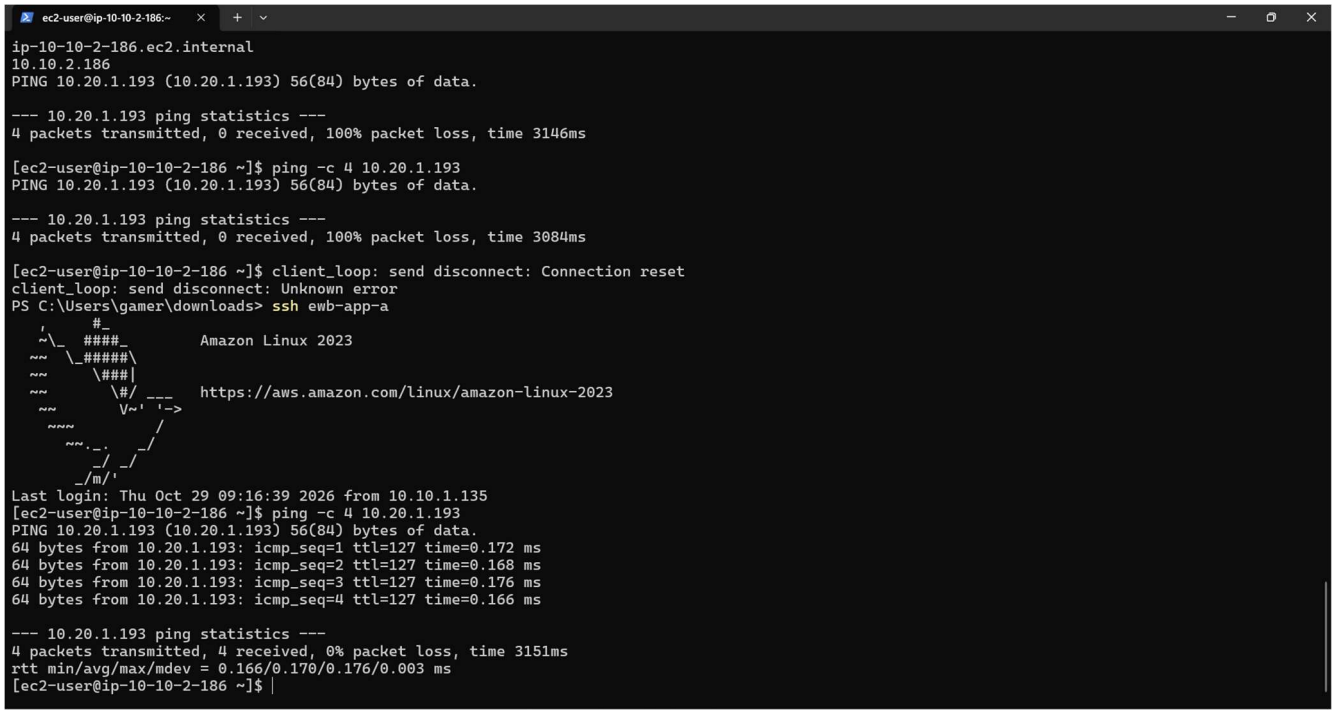
Security groups inventory



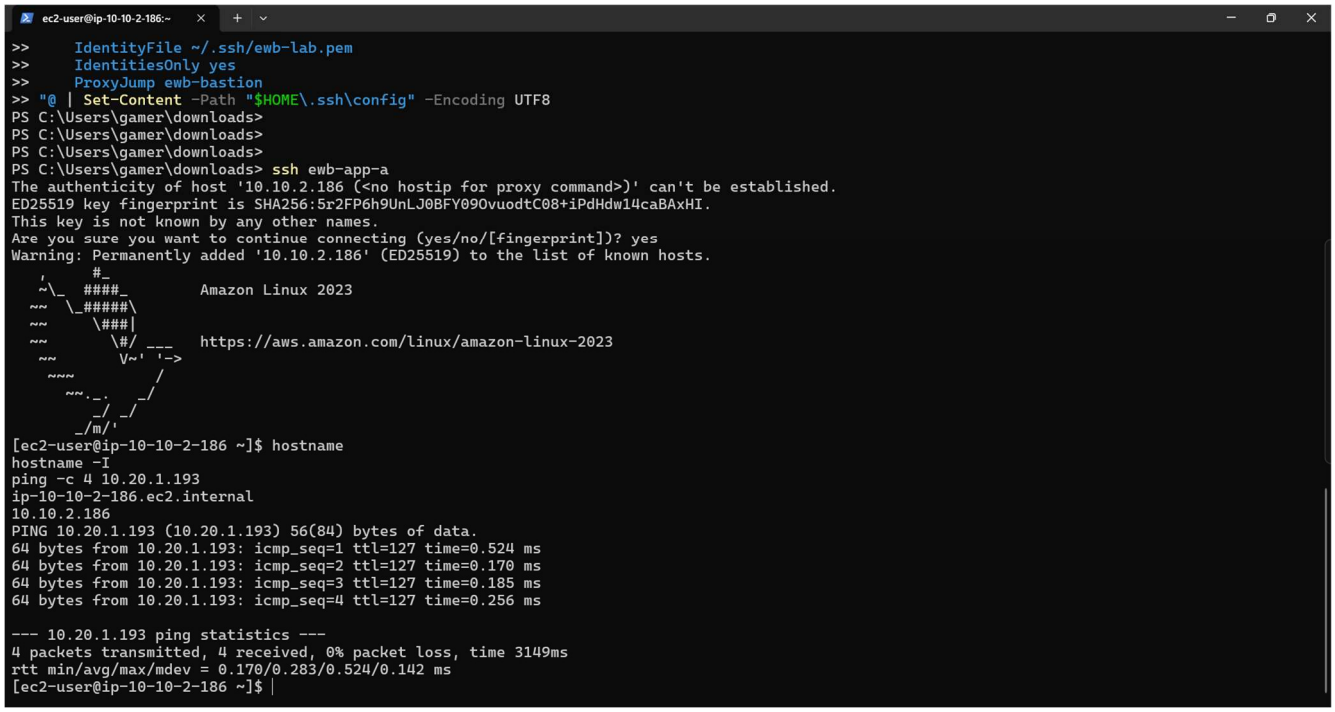
6. Connectivity validation: SSH and ping

The terminal screenshots are arranged to show the recorded baseline failure, connectivity tests, and SSH/ping output. Read the terminal output in sequence; the screenshots are original evidence and have not been altered.

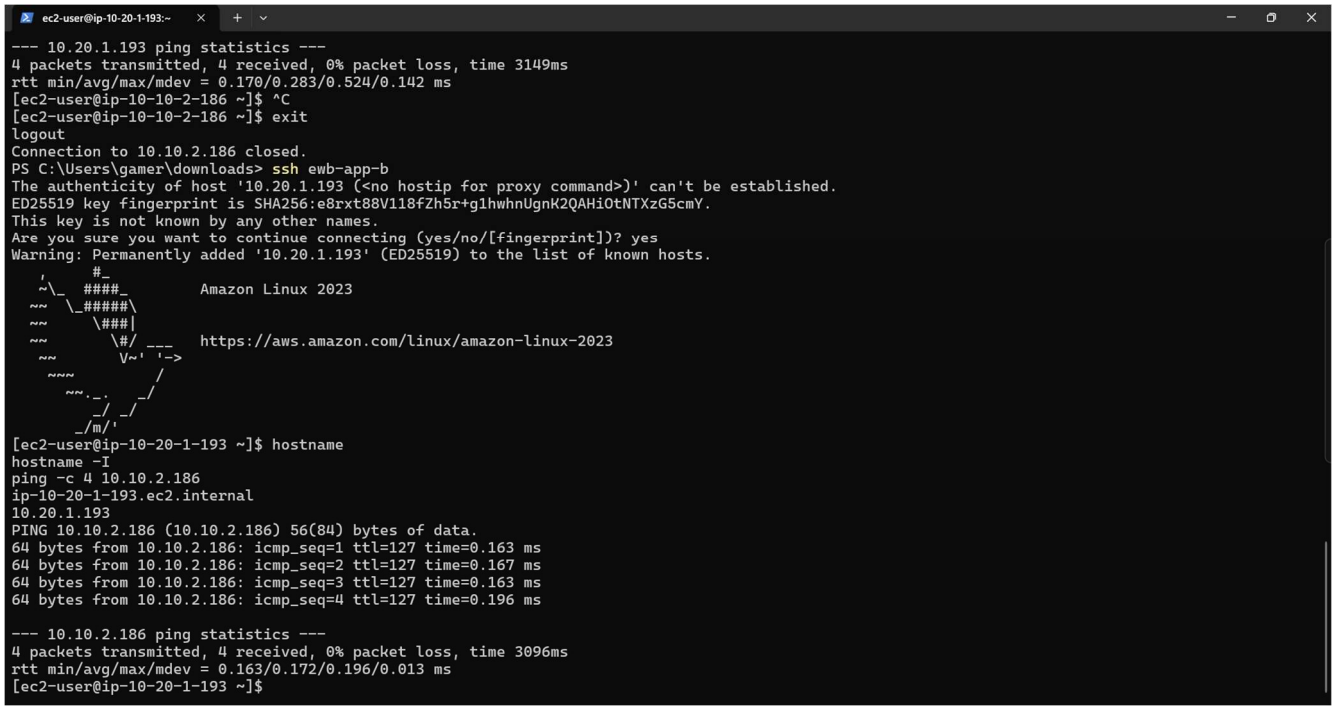
Baseline ping failure / terminal evidence



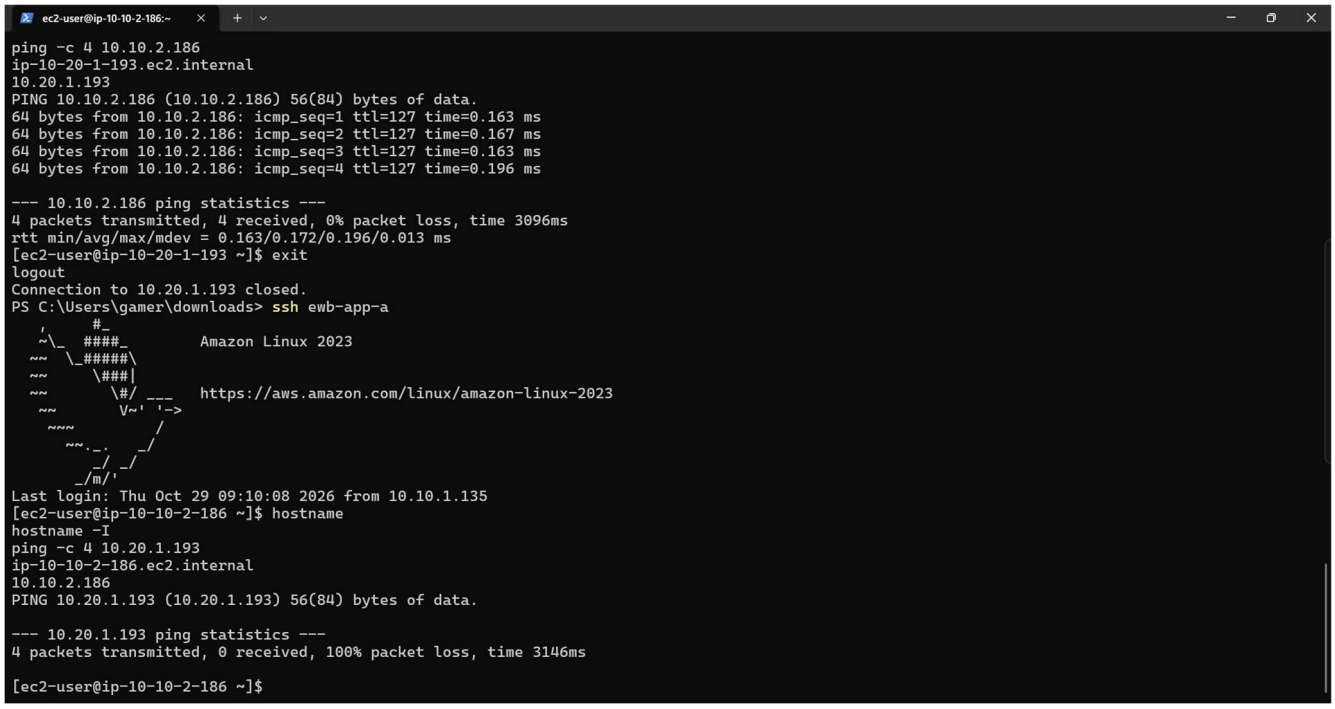
Connectivity and SSH terminal output



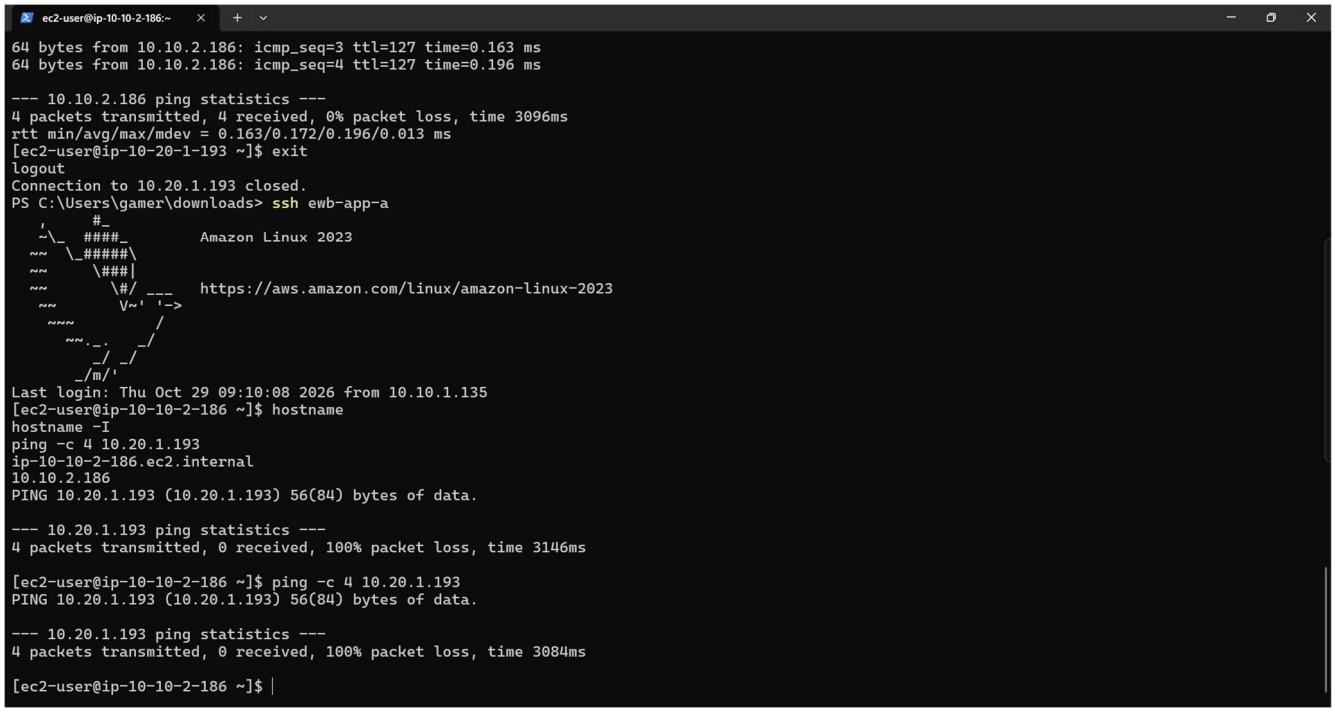
Connectivity test output



Connectivity test output



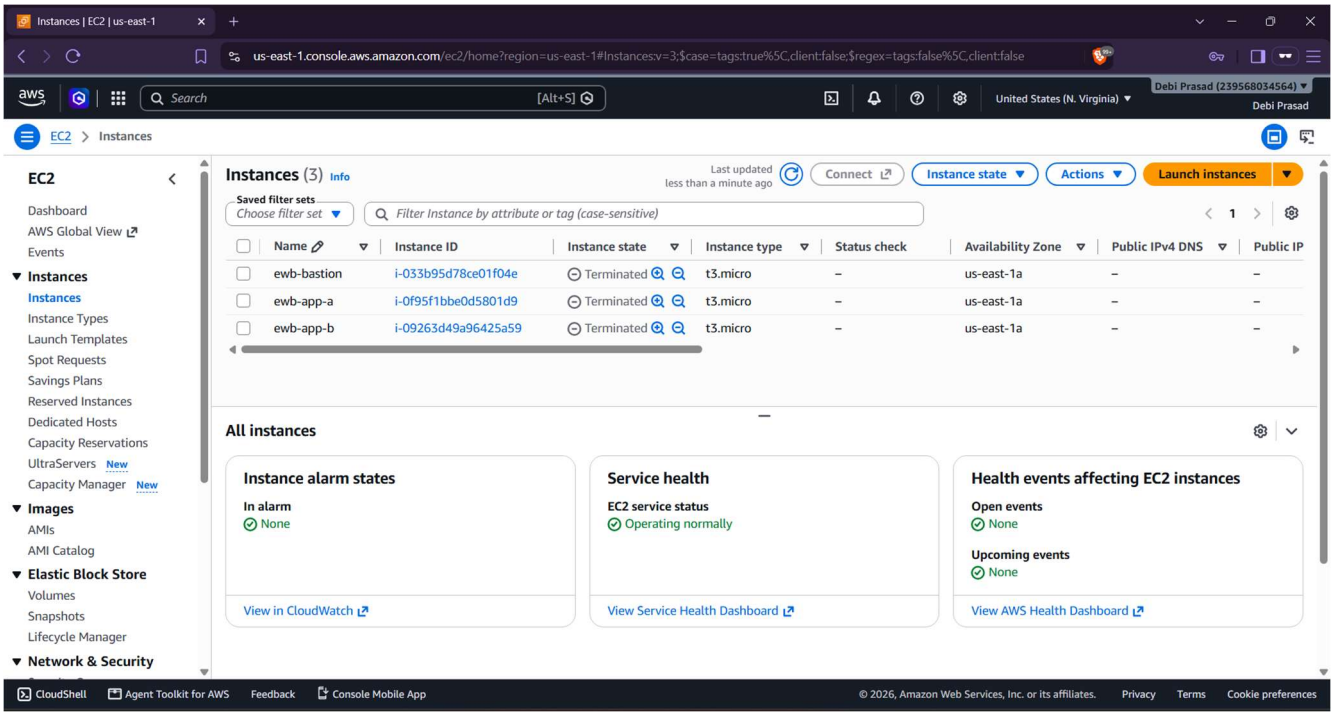
Connectivity test output and failure/recovery history



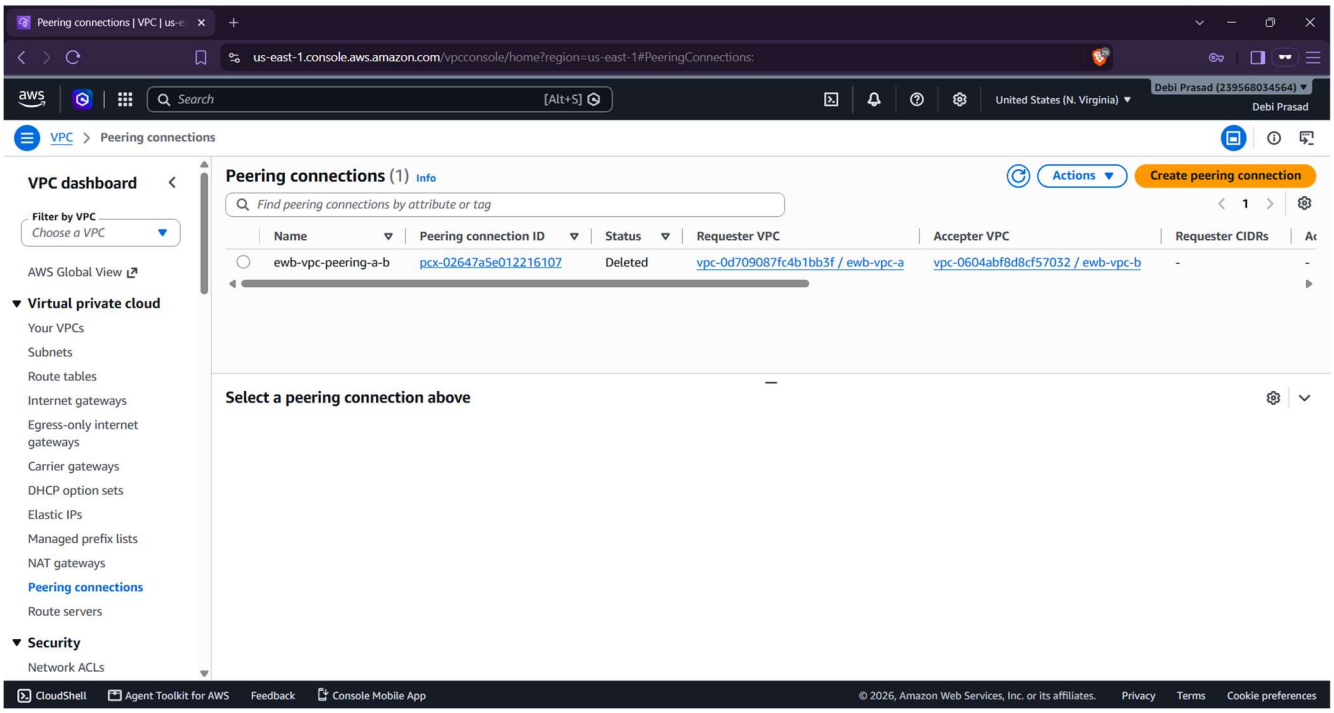
7. Cleanup evidence

These console screenshots record the cleanup actions taken after the lab. They are included as evidence of cleanup, not as evidence of the original active configuration.

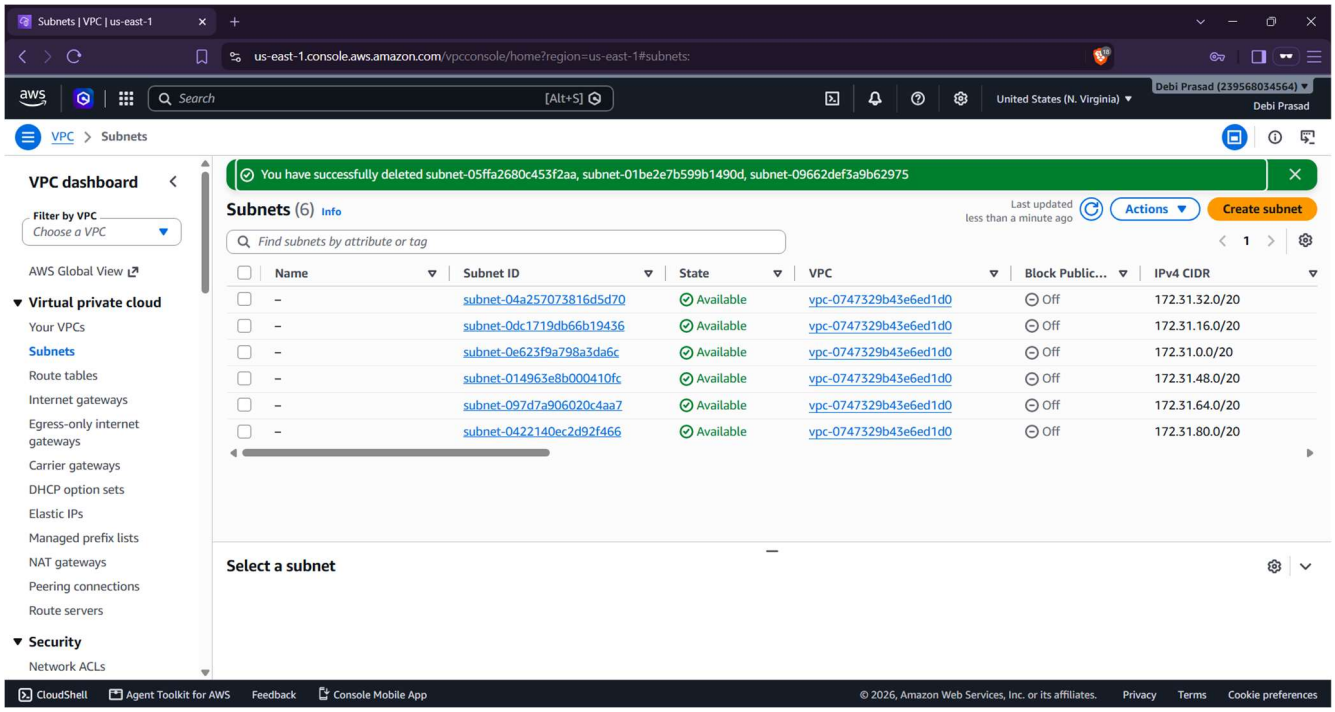
EC2 cleanup / instance status



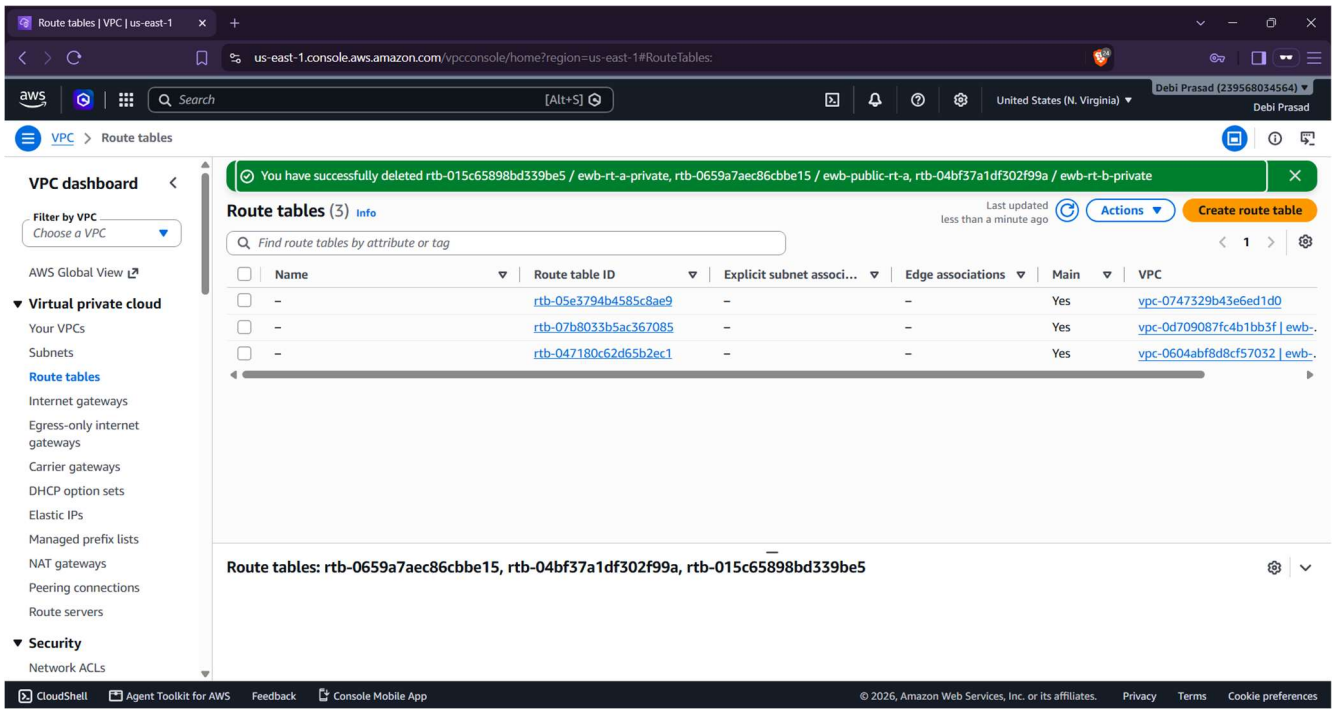
Peering connection cleanup



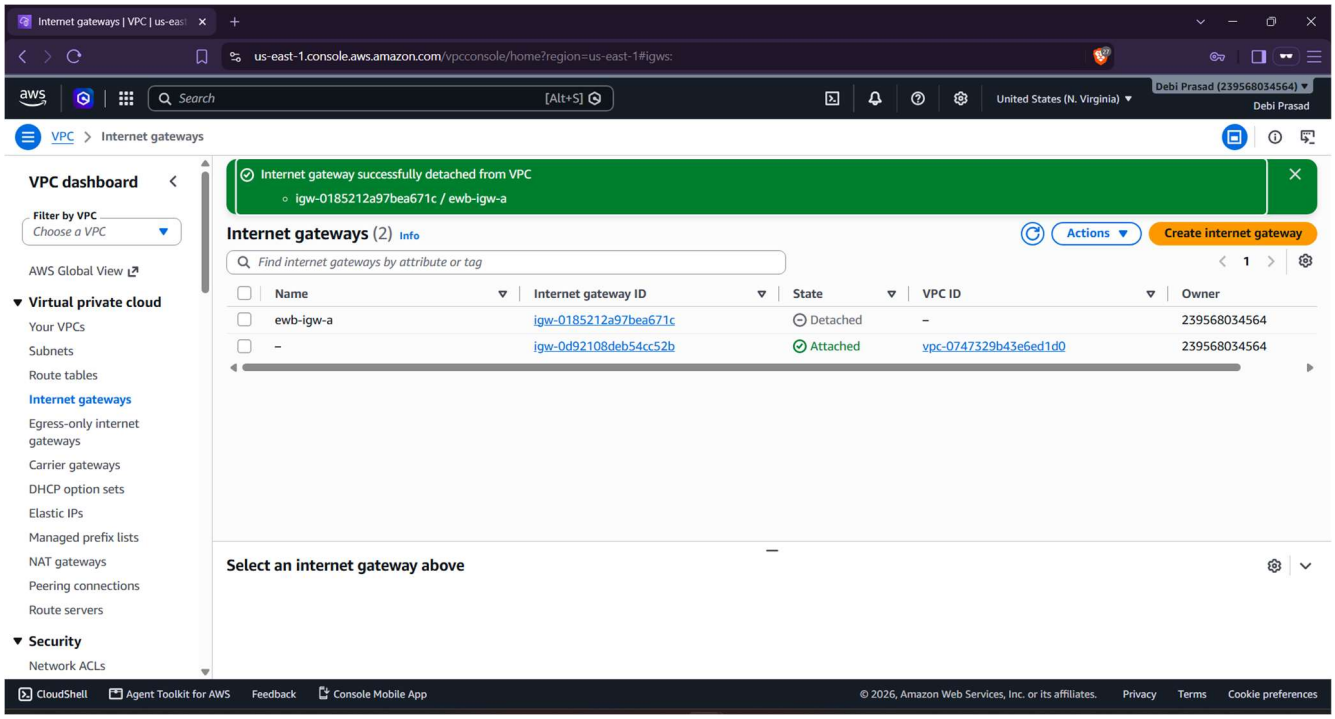
Subnet cleanup



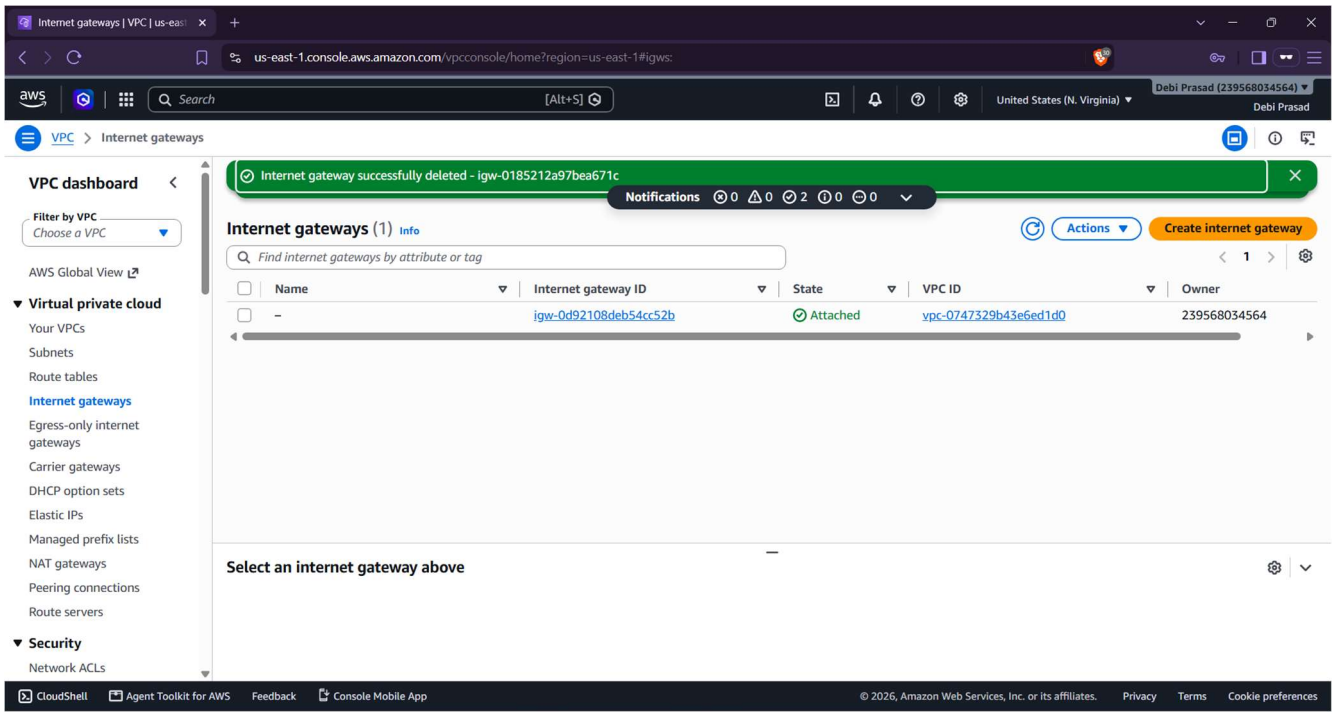
Route-table cleanup



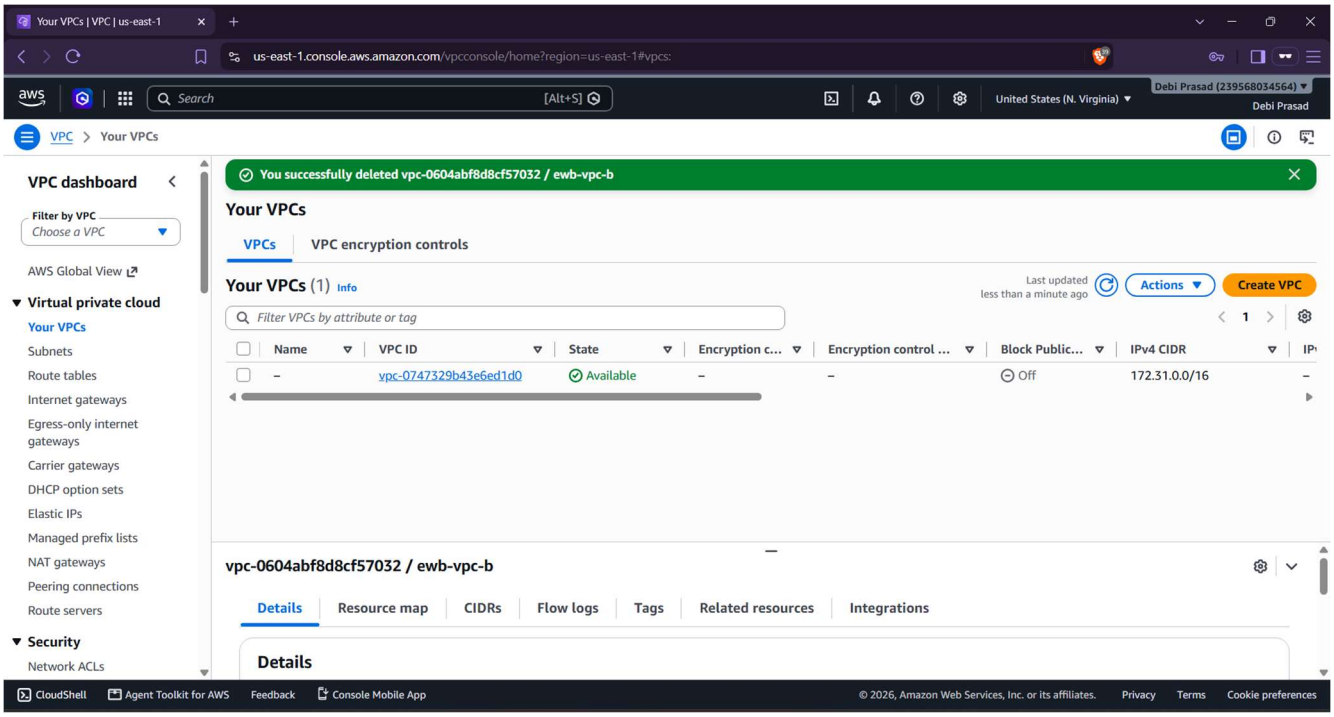
Internet gateway cleanup



Internet gateway deletion confirmation



VPC cleanup



8. Summary and learning outcomes

This lab demonstrates the key design principles of VPC peering: the two VPC CIDR ranges must not overlap; each side needs a route to the other VPC CIDR through the peering connection; security groups must permit the required SSH and ICMP traffic; and private EC2 instances can communicate without public IP addresses when routing and security rules are configured correctly.

The terminal screenshots record SSH sessions and ping tests between the private instances. The captured output includes both successful replies and 100% packet-loss results, so each test should be interpreted according to its displayed result. Cleanup screenshots document removal of the lab resources.

Thank You